

The Netflix logo, consisting of the word "NETFLIX" in a bold, red, sans-serif font, is centered at the top of the slide. The letters are slightly shadowed, giving it a 3D appearance as if it's floating above the black background.

NETFLIX

Capstone Project

Netflix-Movies-and-TV-Shows-

Clustering

Unsupervised Machine Learning

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Problem Statement -

This dataset consists of tv shows and movies available on Netflix as of 2019. The dataset is collected from Flixable which is a third-party Netflix search engine.

In 2018, they released an interesting report which shows that the number of TV shows on Netflix has nearly tripled since 2010. The streaming service's number of movies has decreased by more than 2,000 titles since 2010, while its number of TV shows has nearly tripled. It will be interesting to explore what all other insights can be obtained from the same dataset.

Data Summery

show_id : Unique ID for every Movie / Tv Show

type : Identifier - A Movie or TV Show

title : Title of the Movie / Tv Show

director : Director of the Movie

cast : Actors involved in the movie / show

country : Country where the movie / show was produced

date_added : Date it was added on Netflix

release_year : Actual Releaseyear of the movie / show

rating : TV Rating of the movie / show

duration : Total Duration - in minutes or number of seasons

listed_in : Genere

description: The Summary description

Basic Data Exploration

- The dataset has 7787 observation and 12 features(columns).
- The dataset consist of eleven textual columns and one numerice columns('release_year').
- No duplicate values.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         7787 non-null   object
1   type            7787 non-null   object
2   title           7787 non-null   object
3   director        5398 non-null   object
4   cast            7069 non-null   object
5   country         7280 non-null   object
6   date_added      7777 non-null   object
7   release_year    7787 non-null   int64
8   rating          7780 non-null   object
9   duration        7787 non-null   object
10  listed_in       7787 non-null   object
11  description     7787 non-null   object
dtypes: int64(1), object(11)
memory usage: 730.2+ KB
```

EDA (Checking NaN values)

Null values present in this columns.

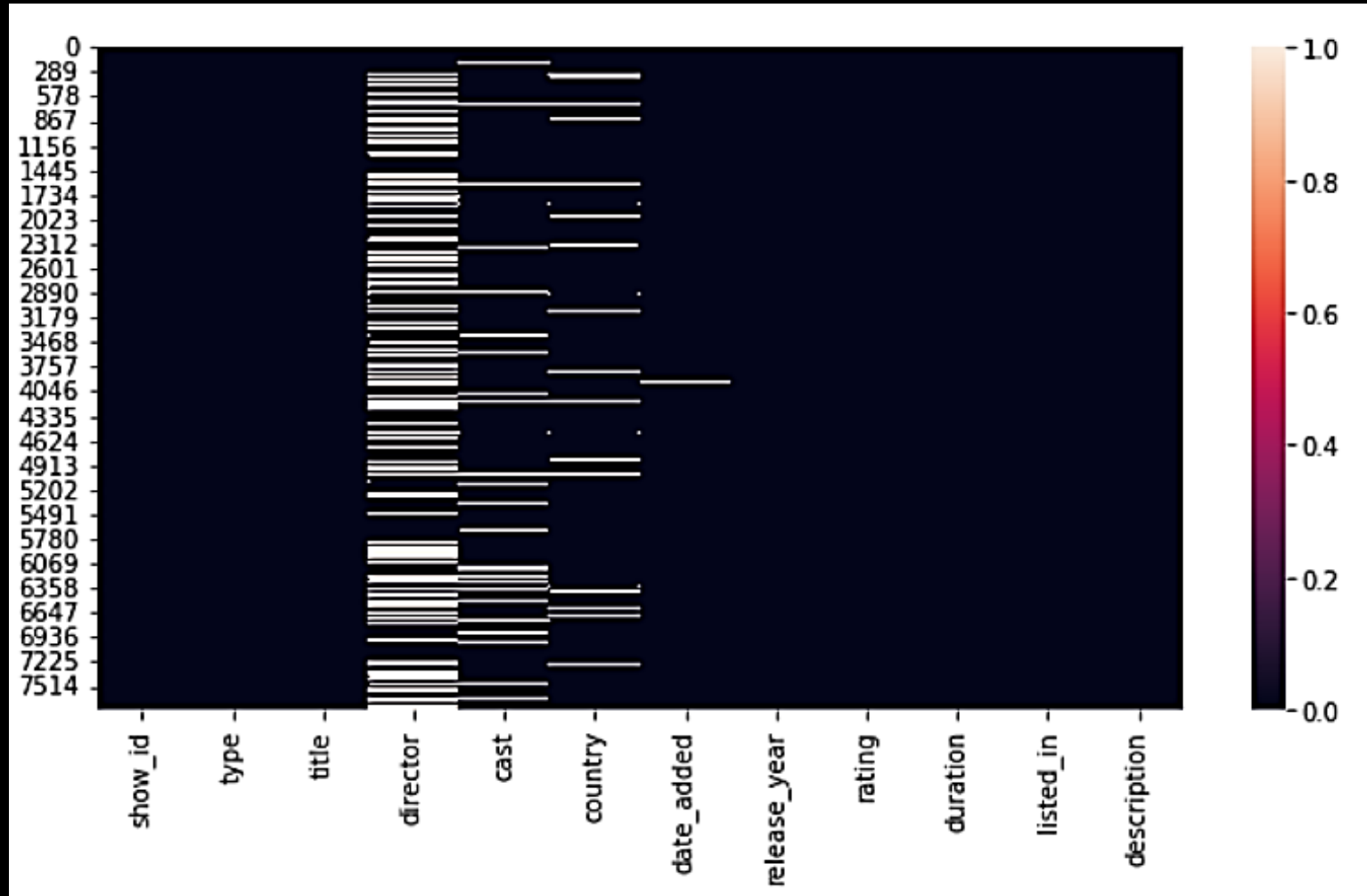
Director

Cast

Country

Date_added

Remaining no missing value present in this columns.



Data Cleaning

director column has highest NaN values 30.7% data is missing

cast column has 9% NaN values

country , **date_added** , **rating** this columns also containing missing values

Production Yearly Growth

Yearly **Movies** release_year

2021 12

2020 411

2019 582

2018 734

2017 744

Yearly **Shows** release_year

2021 19

2020 457

2019 414

2018 386

2017 268

Total **Content** release_year

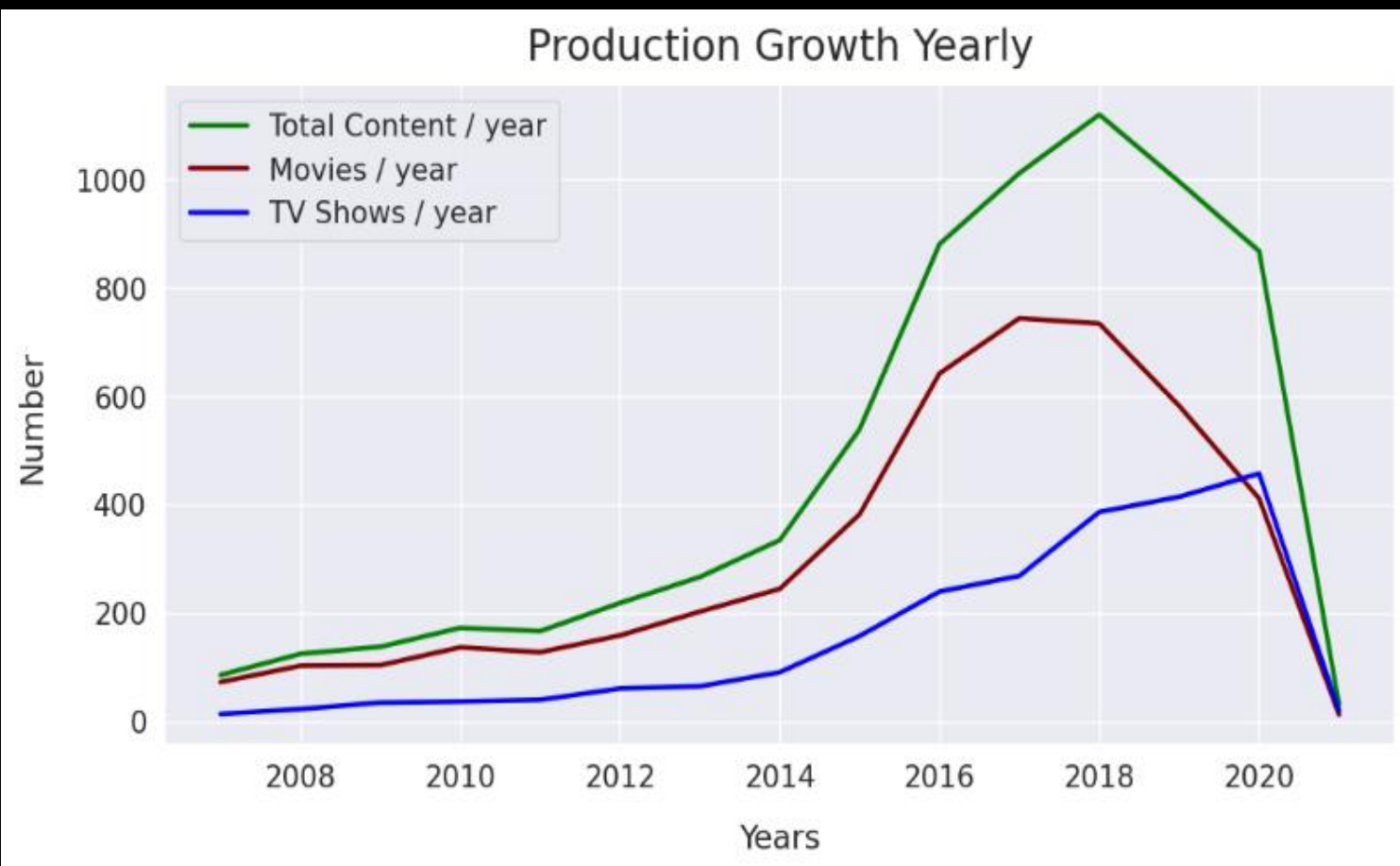
2021 31

2020 868

2019 996

2018 1120

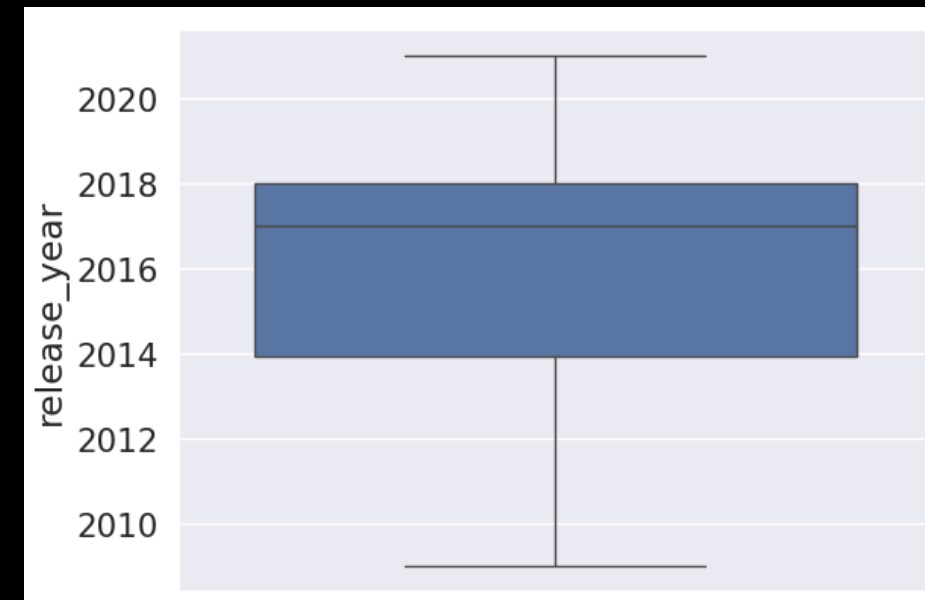
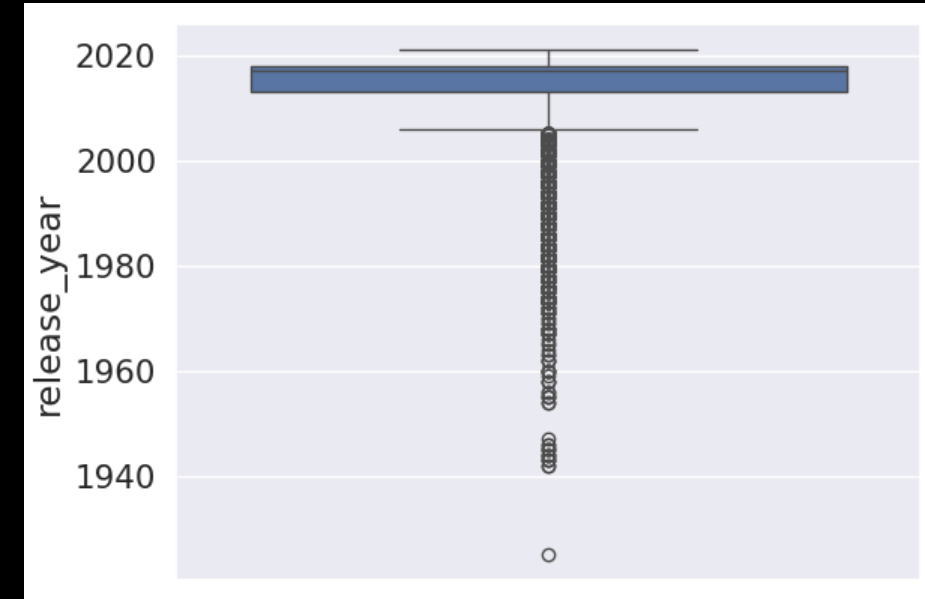
2017 1012



Checking Outliers

As we have seen earlier before 2014 the production growth for Movies & Tv Shows were very less ,that's why it's showing those values(release_year less than 2009) as outliers

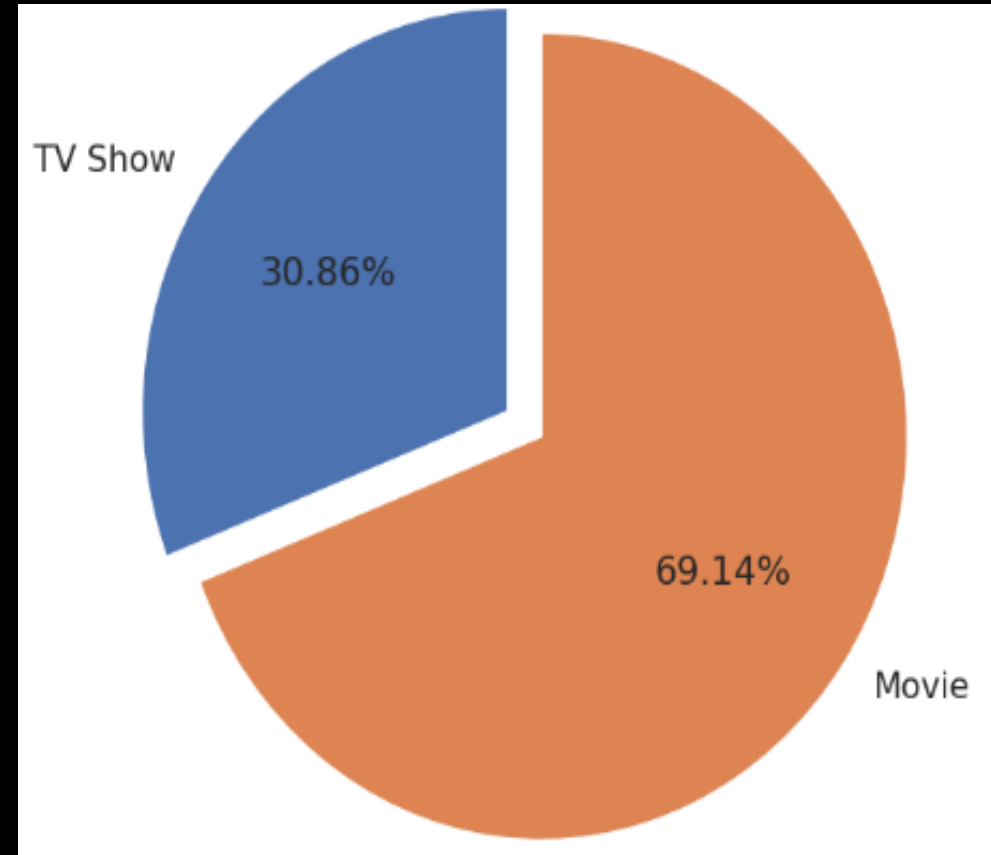
Replacing outliers with mean value using IQR



TV shows or movies

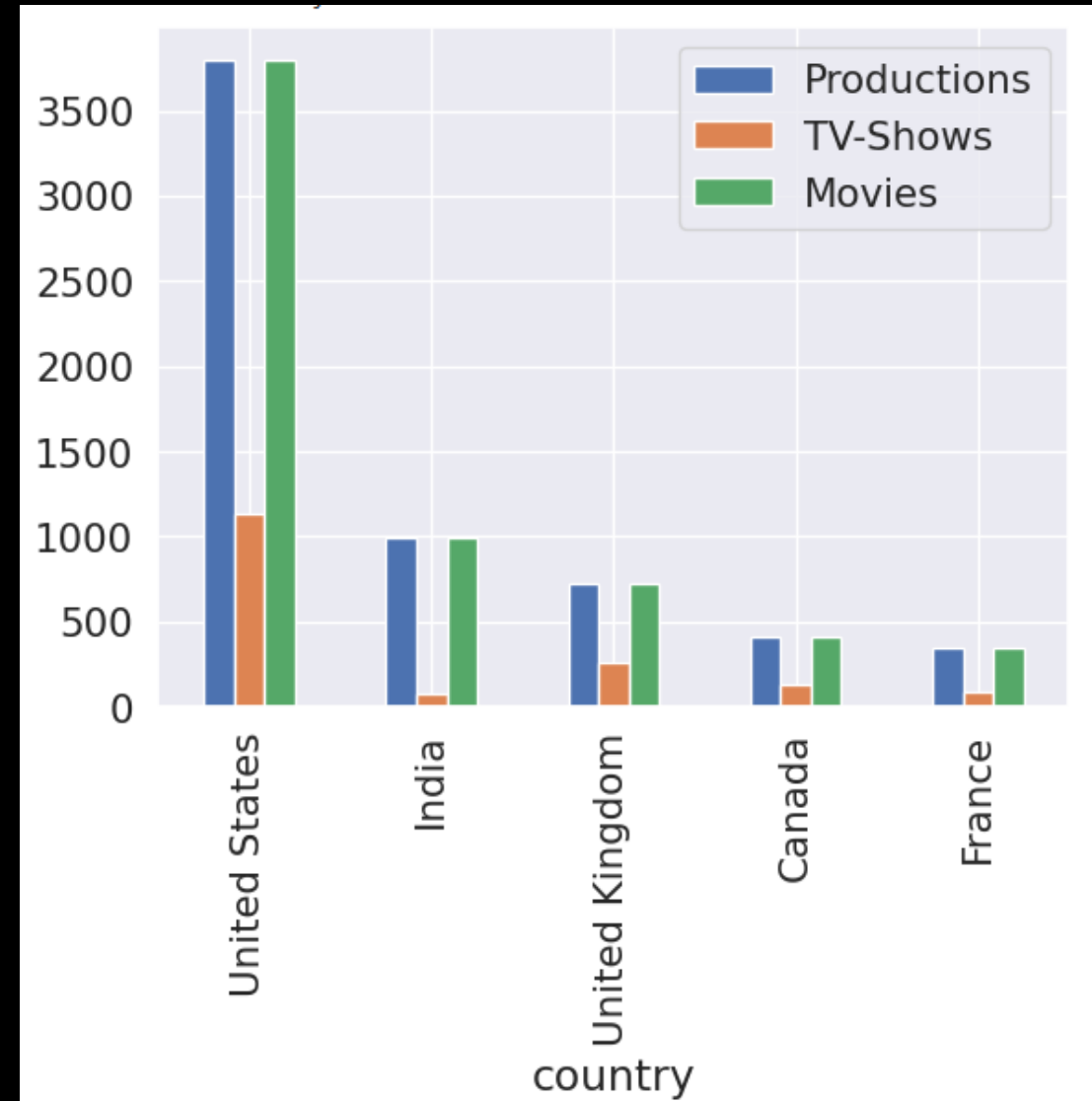
Most of the contents are Movies

Less than $\frac{1}{3}$ content are Tv Shows



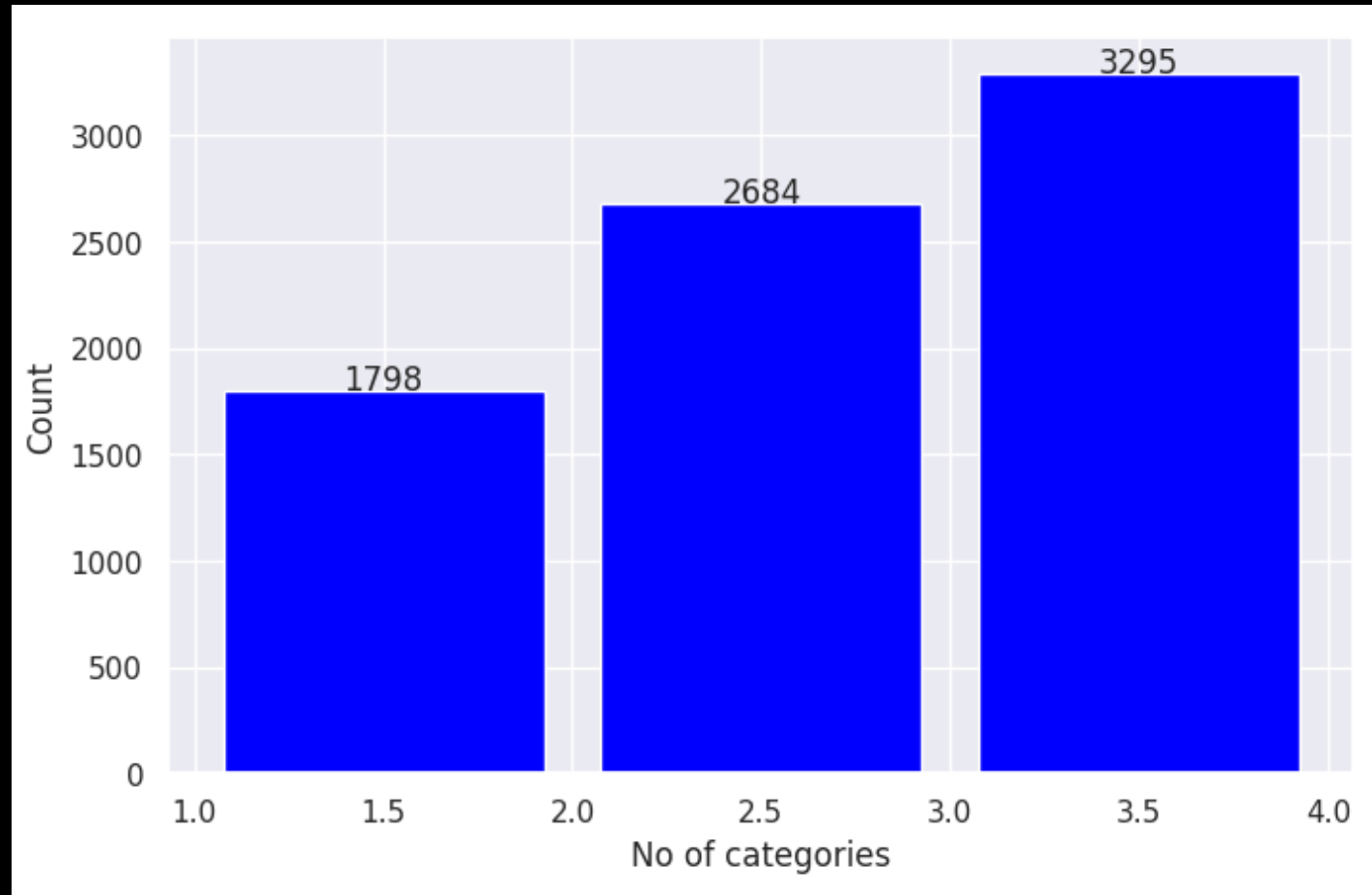
Countries producing most no of contents

United states have the most number of content and then india and so on.
We can conclude that except Japan other countries are producing movies more than TV-shows.

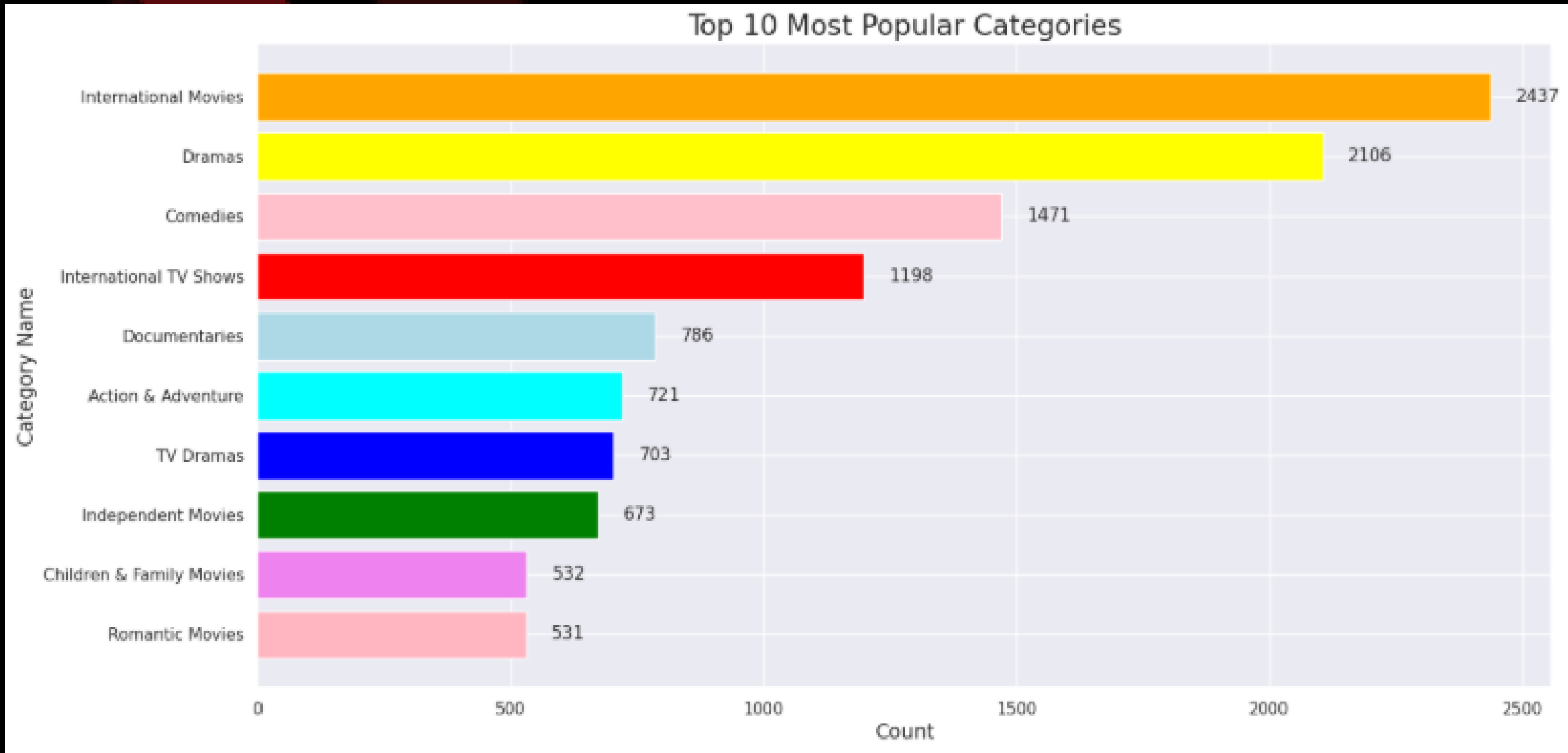


How many on of categories are present there in each content?

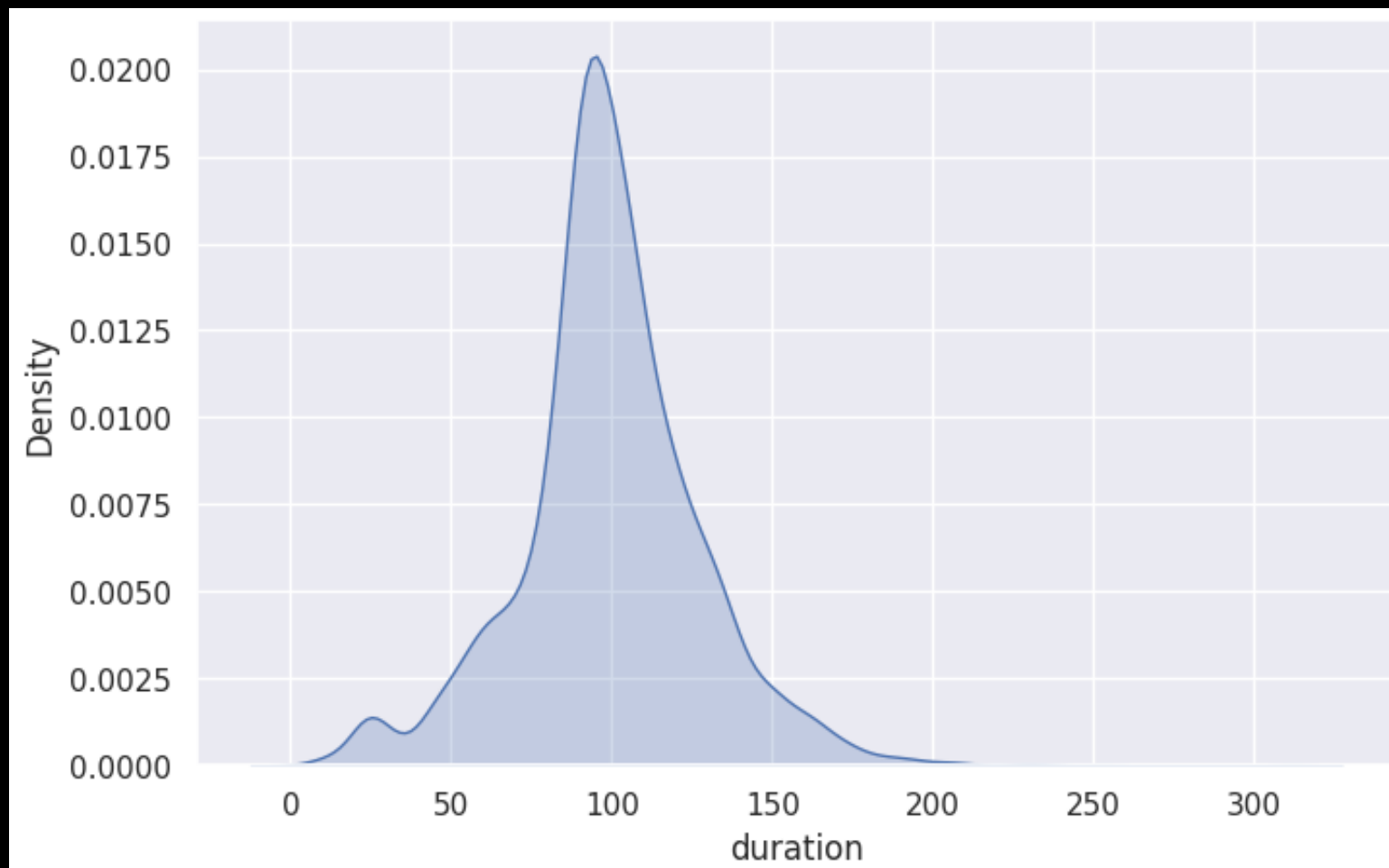
Most of the movies are belonging to **3 categories.**



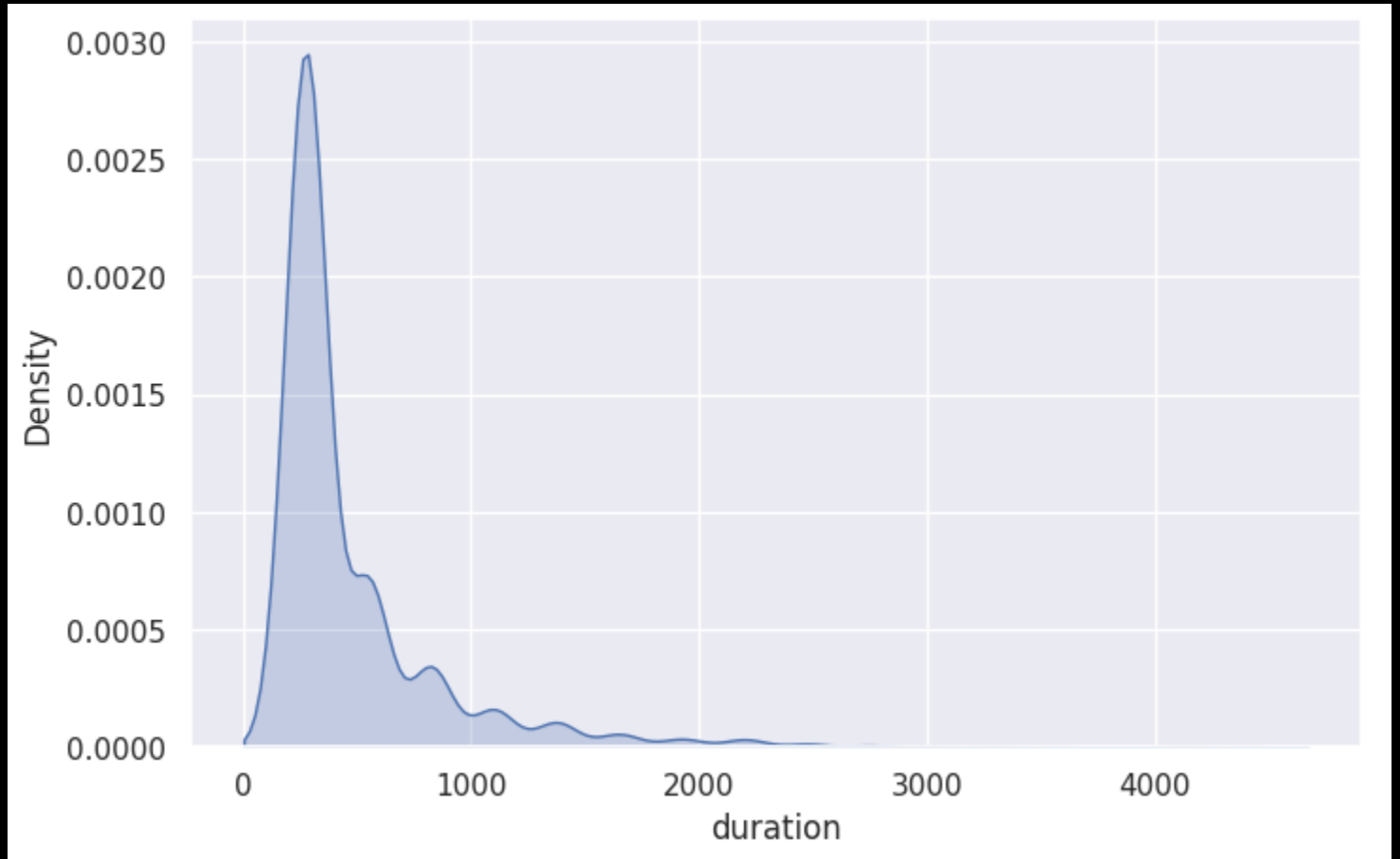
TOP 10 Most Occurred Category By Count



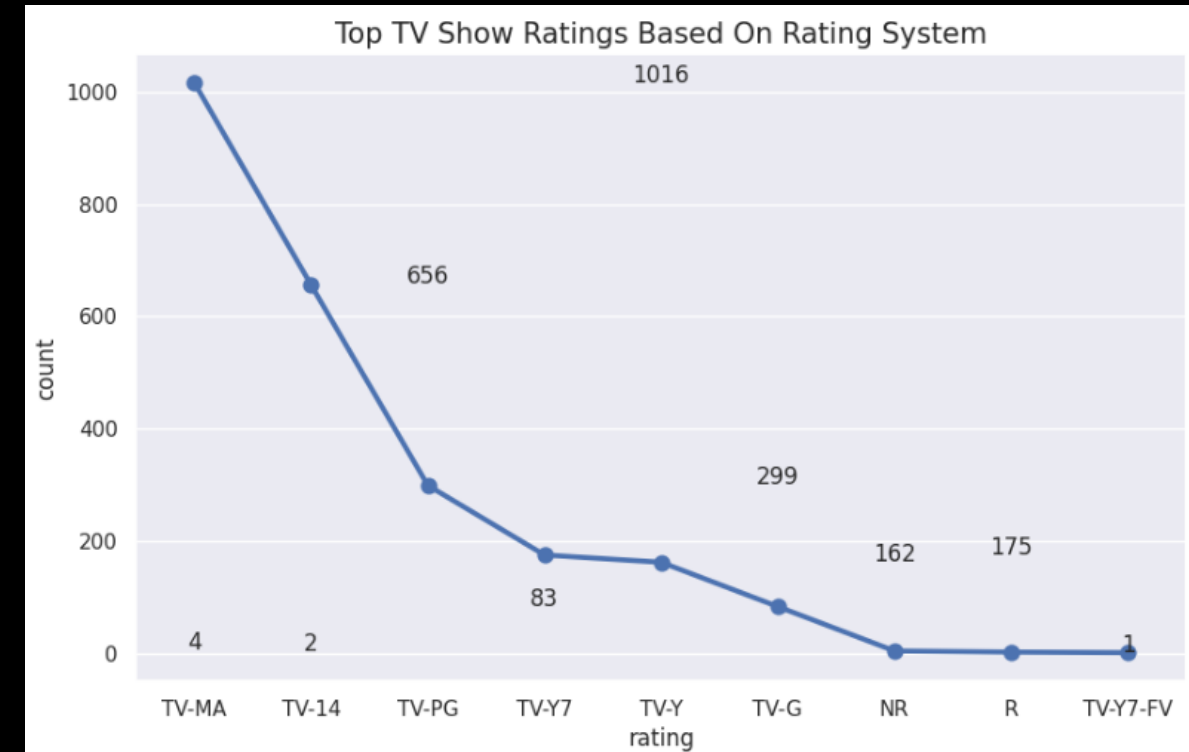
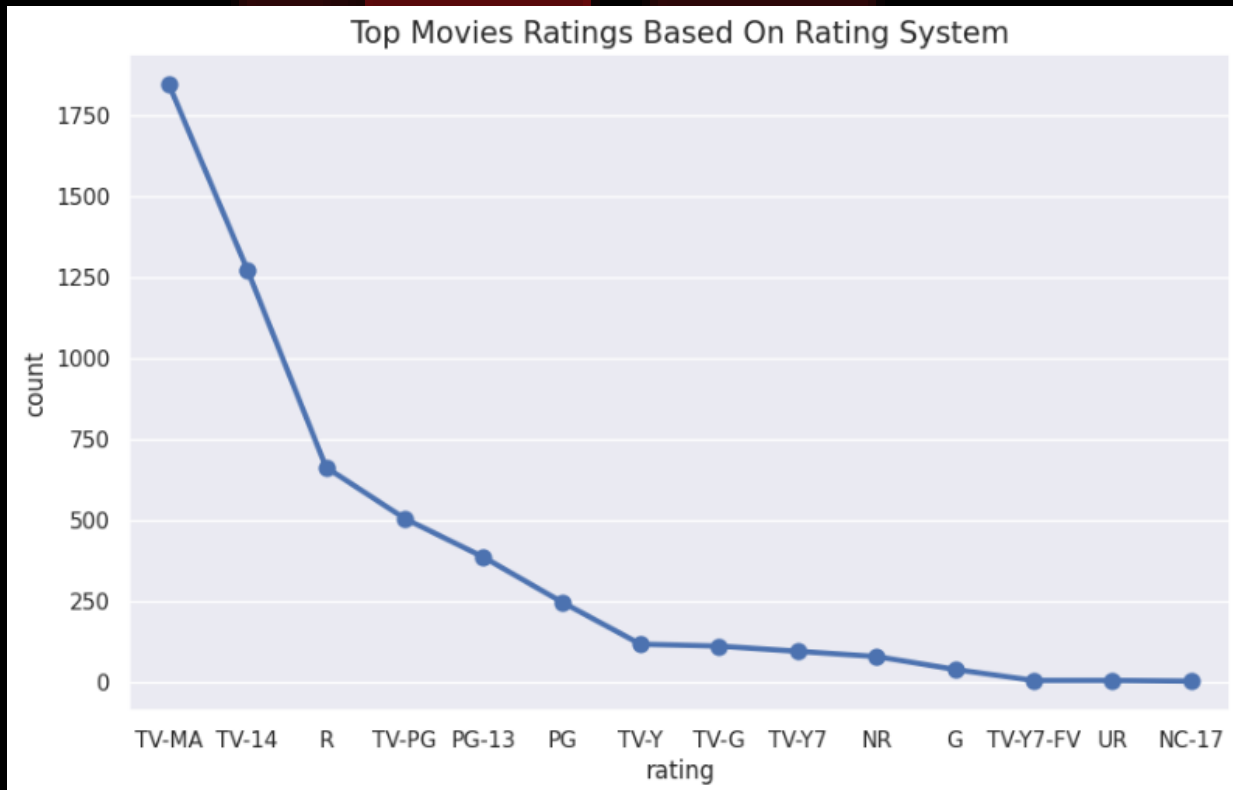
Movies wise density plot



TV-Shows wise density plot



Top Content Based on Rating



Most of the contents got ratings like
TV-MA (For Mature Audiences)
TV-14 (May be unsuitable for children under 14)
TV-PG (Parental Guidance Suggested)
NR (Not Rated)

Word Cloud

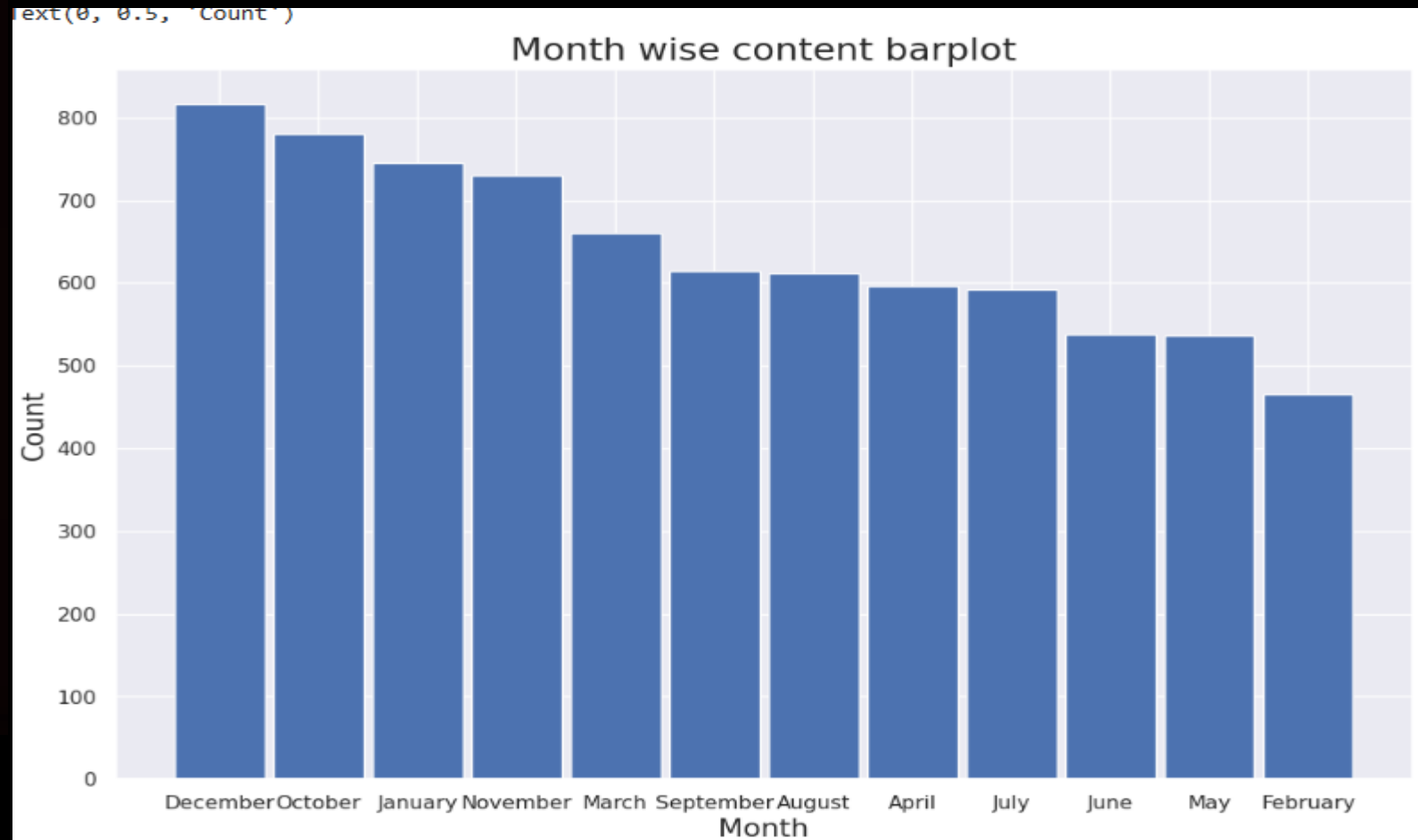
A word cloud (also known as a **tag cloud**) is a visual representation of words. Cloud creators are used to **highlight popular words and phrases** on frequency and relevance. They provide you with **quick and simple visual insight** that can lead to more in depth analyses.

It seems like words like "Love", "Man", "World", "Story", "Christmas" are very common in titles.

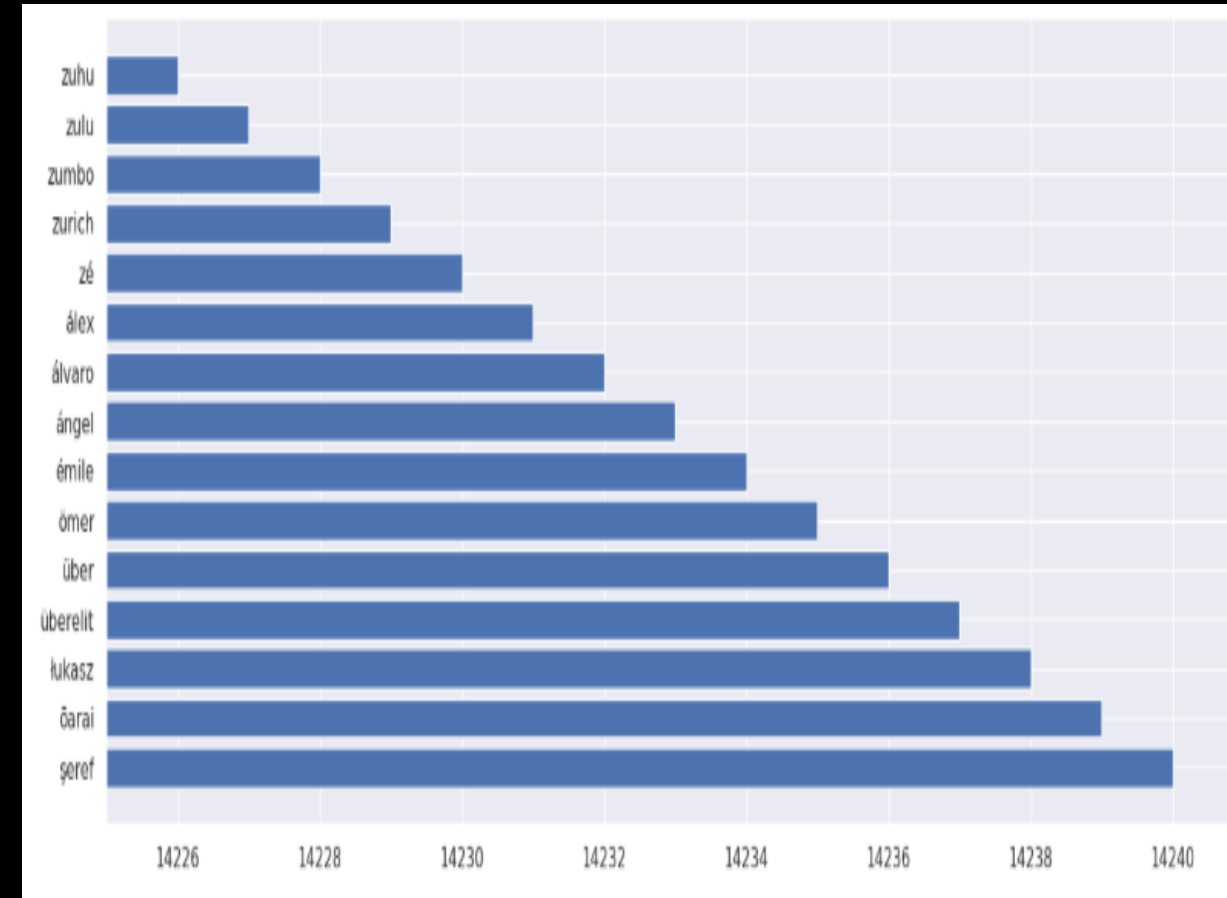
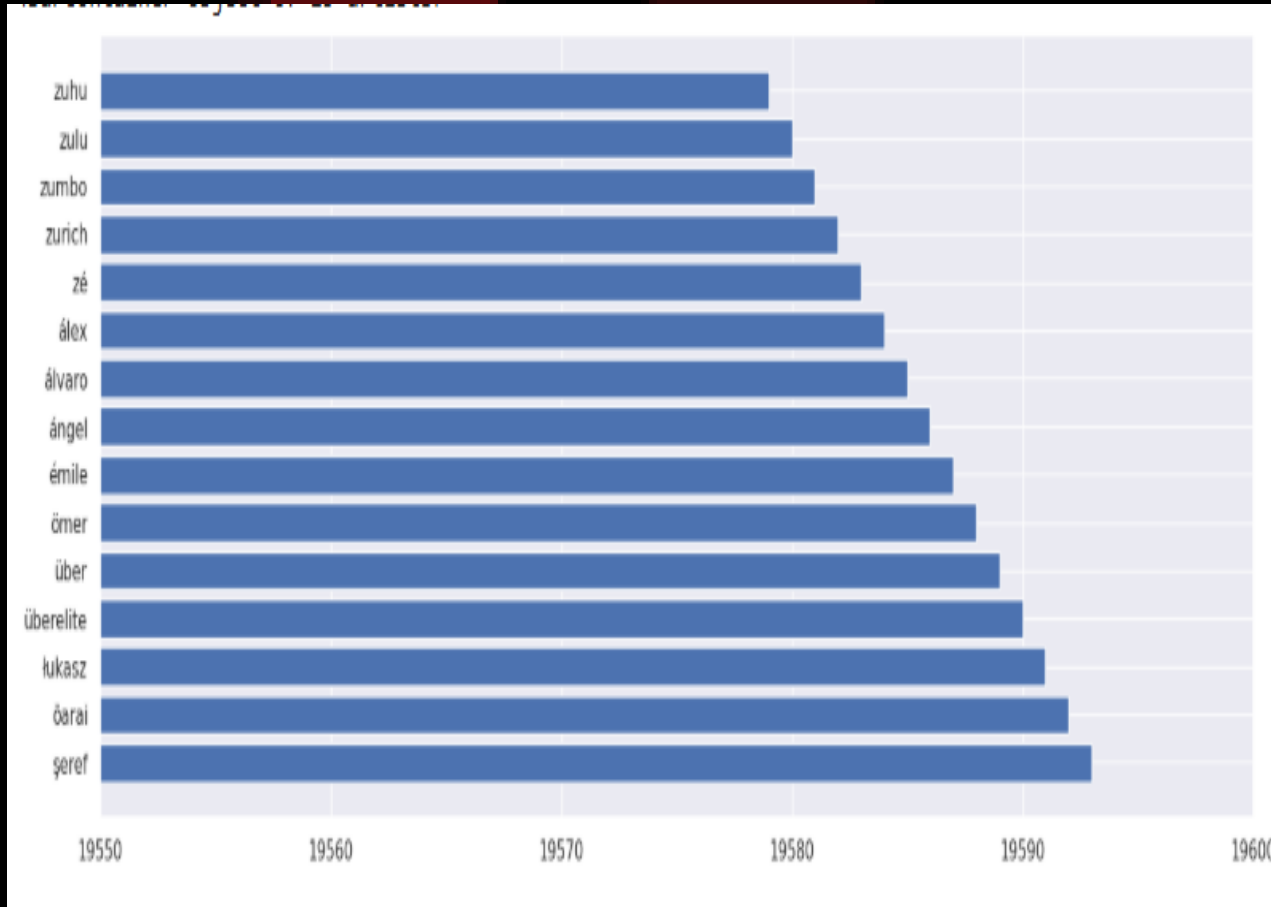


Barplot based on release month

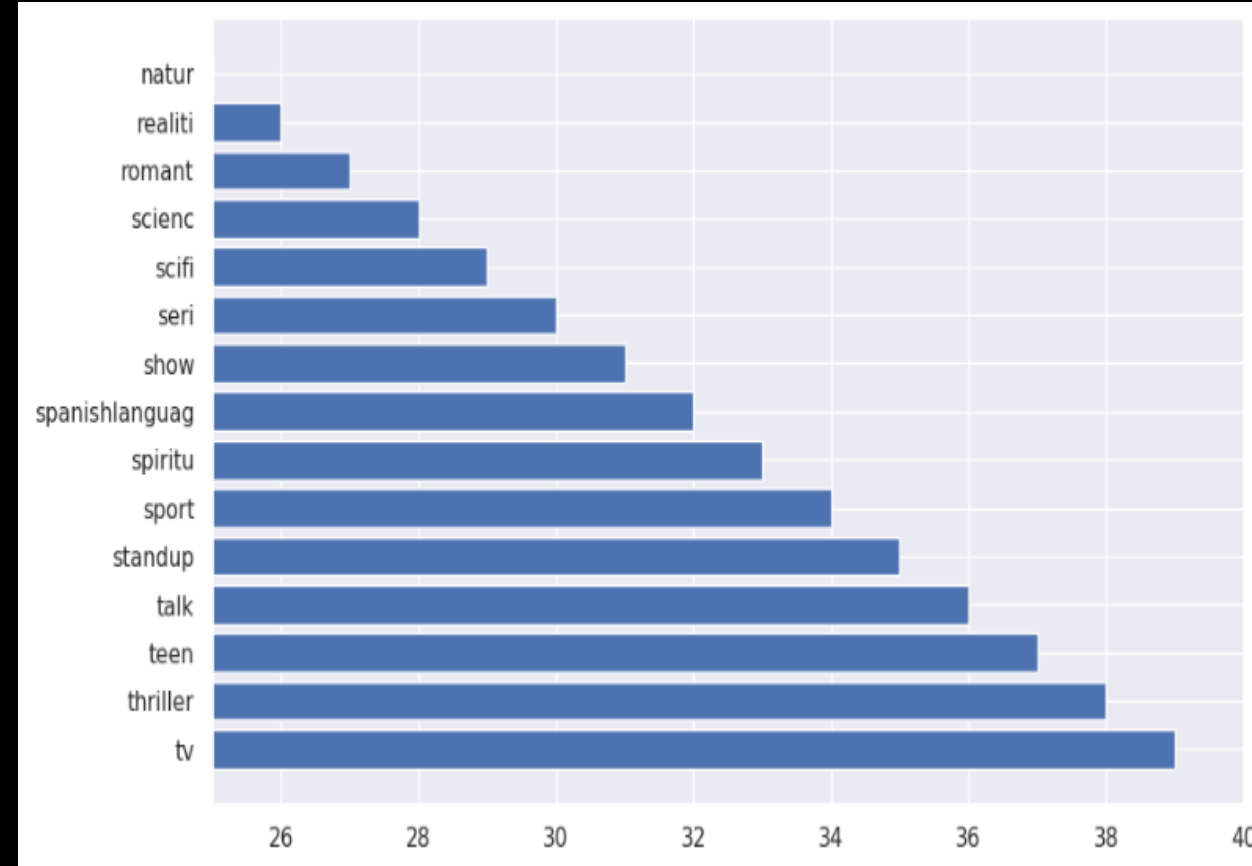
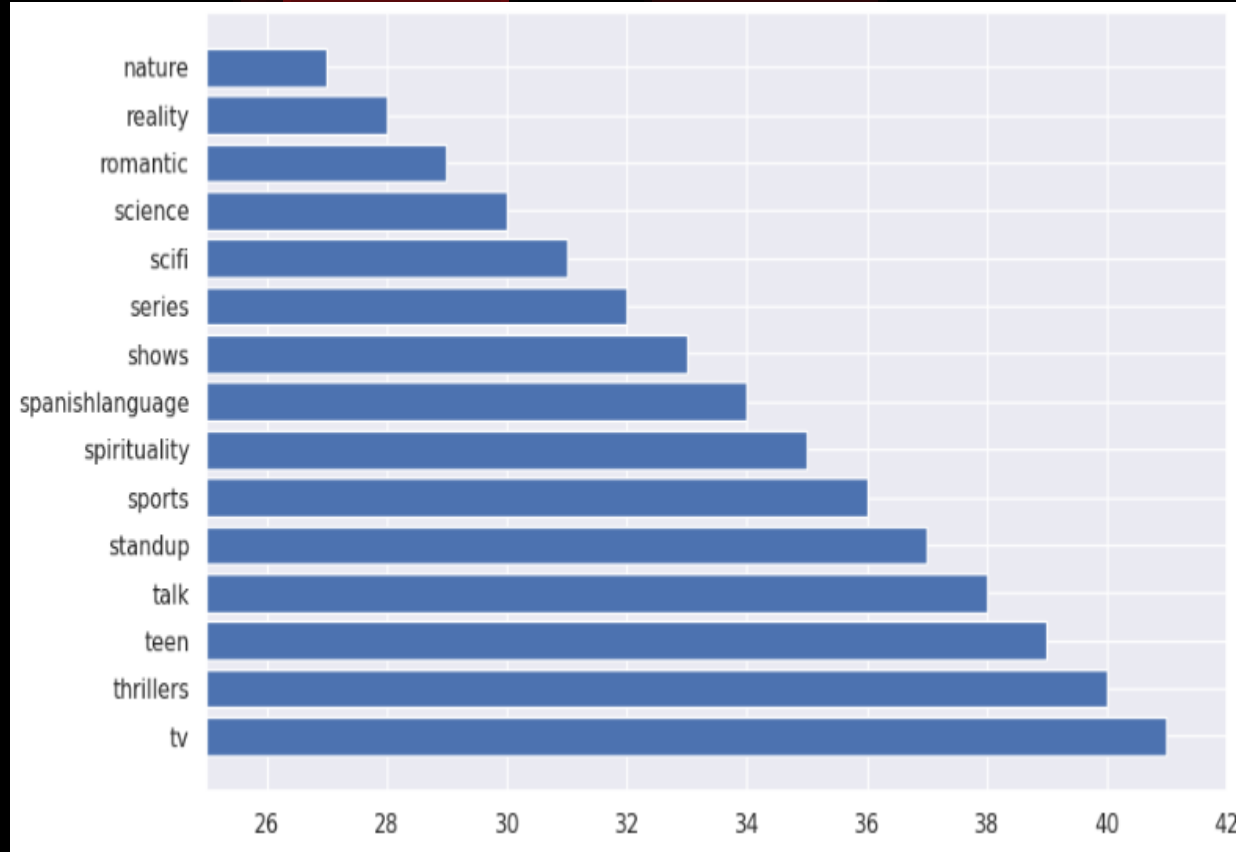
We can say that **December** is the **holiday** season and it also has **Christmas**, so in that month most of the content got uploaded.



Before & after streaming most occurred words in description



Before & after streaming most occurred words in listed-in



Feature selection & ML algo used

Only selected 3 feature, to do clustering

No_of category

Length(description)

Length(listed-in)

Using StandardScaler

Used 5 algo to find out best k value

1. Silhouette score
2. Elbow Method
3. DBSCAN
4. Dendrogram
5. AgglomerativeClustering

1. Silhouette score

Silhouette Coefficient Formula

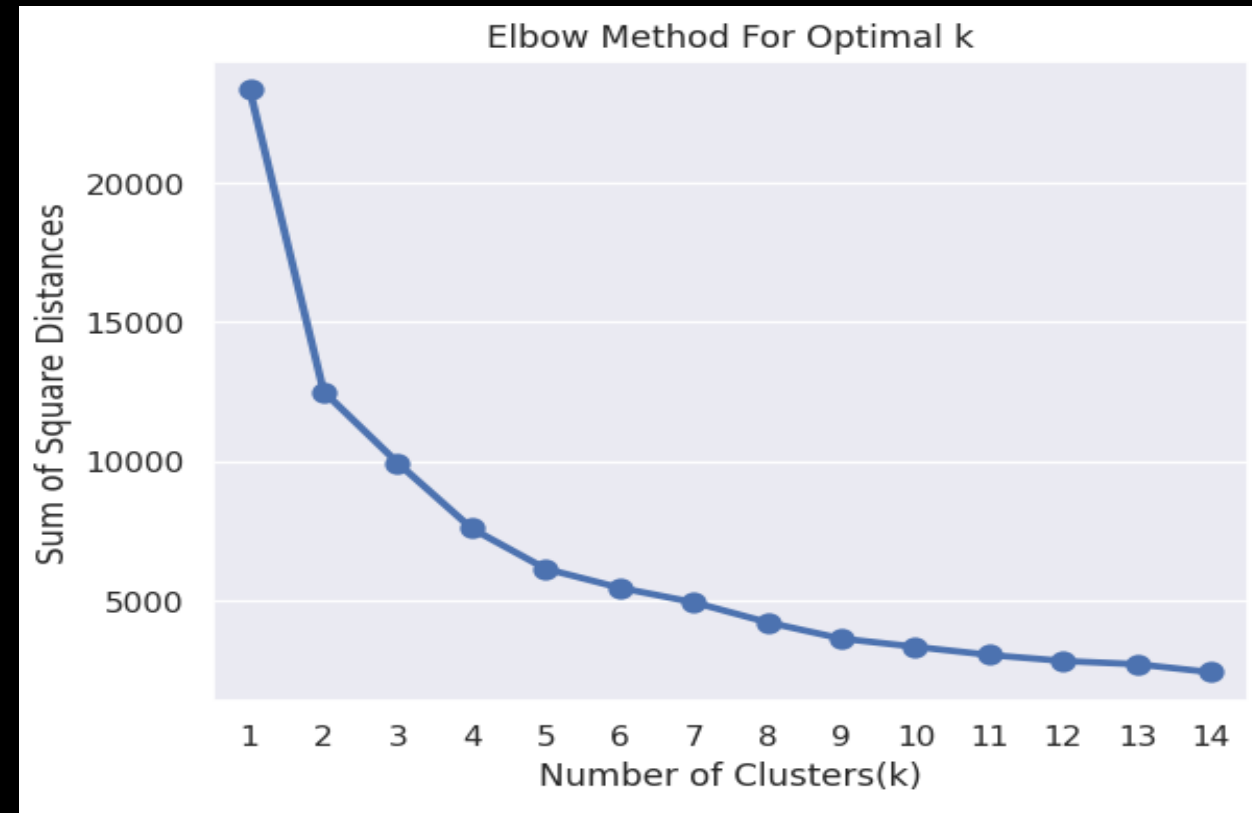
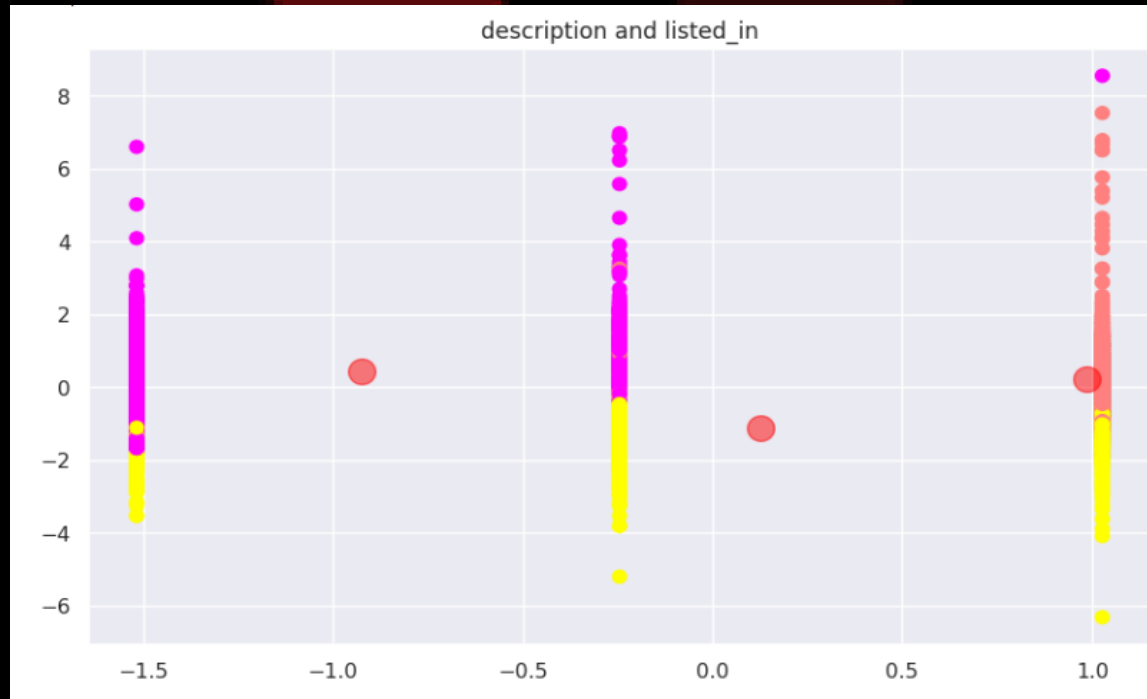
$$S = \frac{(b-a)}{\max(a,b)}$$

- **mean intra-cluster distance (a)** :- Mean distance between the observation and all other data points in the same cluster.
- **mean nearest-cluster distance (b)** :- Mean distance between the observation and all other data points of the next nearest cluster. This distance can also be called a.

The value of the silhouette coefficient is between [-1, 1]. A score of 1 denotes the best meaning that the data point i is very compact within the cluster to which it belongs and far away from the other clusters. The worst value is -1. Values near 0 denote overlapping clusters

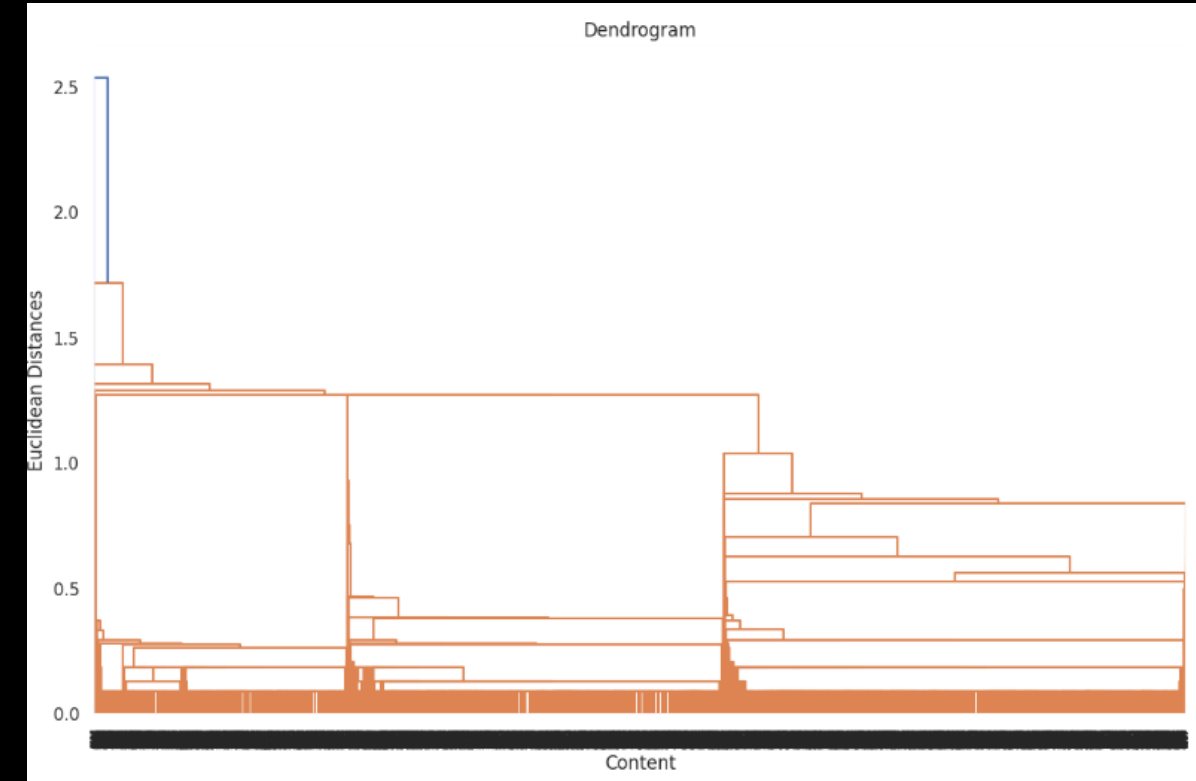
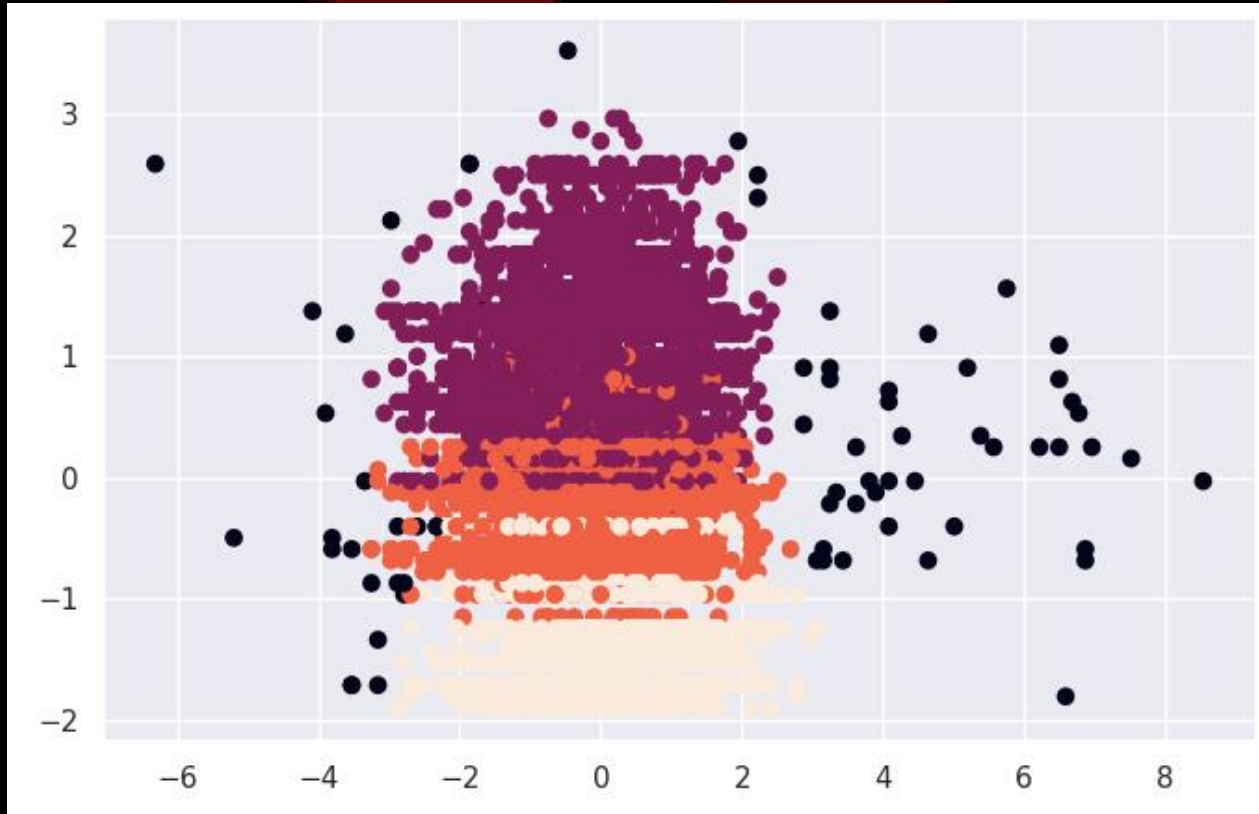
	n clusters	silhouette score
0	2	0.428
3	5	0.371
5	7	0.368
4	6	0.368
8	10	0.364
9	11	0.363
7	9	0.362
6	8	0.357
10	12	0.357
2	4	0.354
1	3	0.351
12	14	0.348
13	15	0.347
11	13	0.338

2. Elbow Method



The elbow method runs **k means clustering** on the data-set for a range of values for **k** (say from 1-15) and then for each value of **k** computes **WCSS** value. By default, the distortion score is computed, the sum of square distance from each point to its assigned center.

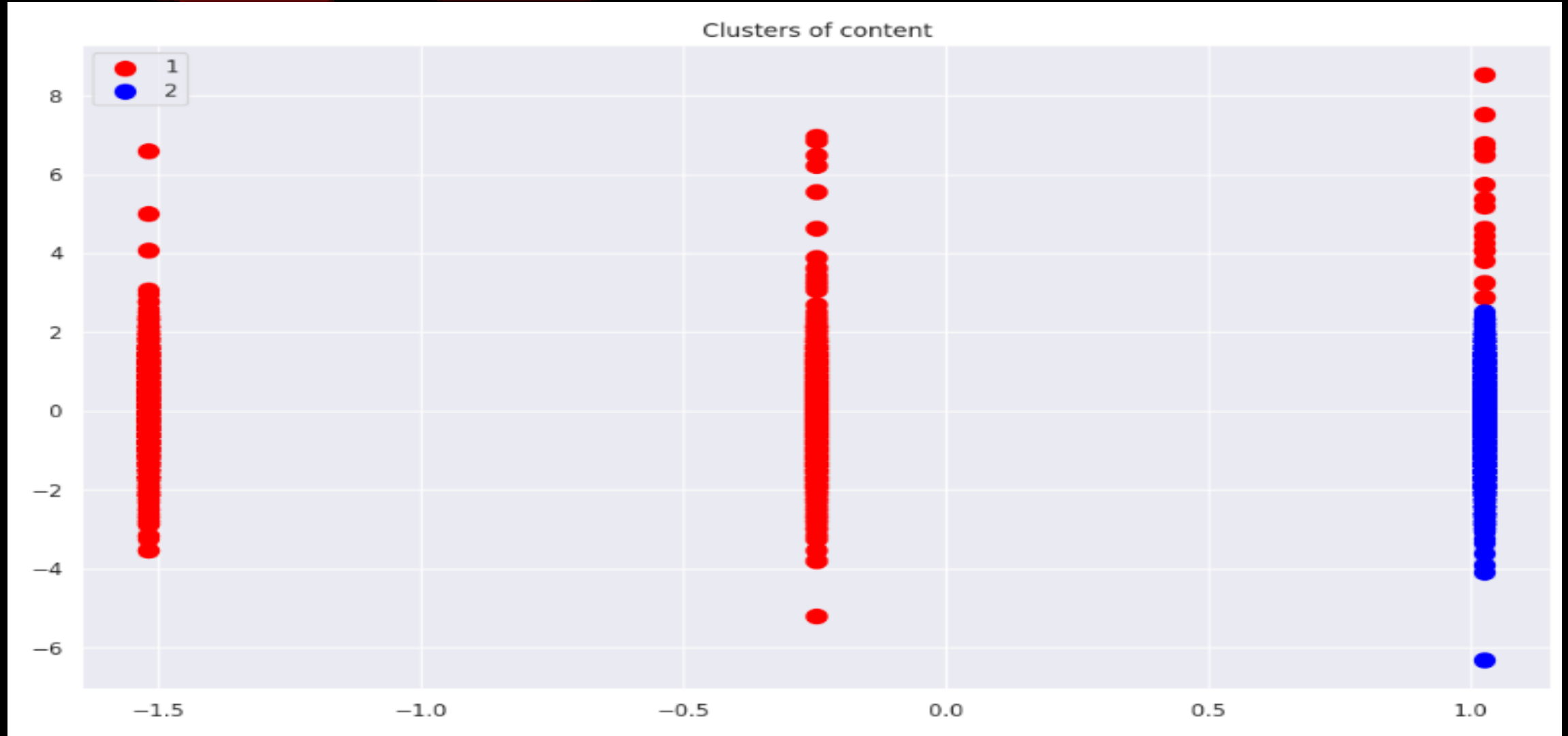
3&4 DBSCAN & Dendrogram



The black colour dots(*) are noise

The number of clusters will be the number of vertical lines which are being intersected by the line drawn using the threshold

5 AgglomerativeClustering



Conclusion

- Director and cast contains a large number of null values so we will drop these 2 columns.
- In this dataset there are two types of contents where 30.86% includes TV shows and the remaining 69.14% carries Movies.
- We have reached a conclusion from our analysis from the content added over years that Netflix is focusing movies and TV shows (From 2016 data we get to know that Movies is increased by 80% and TV shows is increased by 73% compare)
- From the dataset insights we can conclude that the greatest number of TV Shows released in 2017 and for Movies it is 2020
- On Netflix USA has the largest number of contents. And most of the countries preferred to produce movies more than TV shows.
- Most of the movies are belonging to 3 categories TOP 3 content categories are International movies , dramas , comedies.
- In text analysis (NLP) I used stop words, removed punctuations , stemming & TF-IDF vectorizer and other functions of NLP.
- Applied different clustering models like Kmeans, hierarchical, Agglomerative clustering, DBSCAN on data we got the best cluster arrangements.
- By applying different clustering algorithms to our dataset .we get the optimal number of cluster is equal to 3